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Summary

I'm a PhD candidate at the VR Lab of Beihang University from 2021, supervised by Prof. Qinping Zhao and Prof. Yang Gao. My research focuses on physical simulation in computer graphics. Currently I am a visiting student at University College London, supervised by Prof. He Wang. I am about to graduate in 2026/2027. Interest: Numerical Methods, Physical Simulation, Physical Based Animation, Reinforcement Learning, Computer Graphics, AI4S, PDE.

Education

University College London, Visiting PhD in Computer Science — Sept 2025 – Sept 2026

Beihang University (BUAA), PhD in Computer Science — Sept 2021 – Dec 2026

Beihang University (BUAA), MSc in Power Engineering and Engineering Thermophysics — Sept 2018 – June 2021

University of Michigan, Dearborn, Exchange Student in Energy and Power Engineering — Sept 2016 – May 2017

North China Electric Power University (Beijing), BSc in Energy and Power Engineering — Sept 2014 – July 2018

Publications

RLMuscle: Muscle-Driven Character Animation via Reinforcement Learning — Under Review
Chunlei Li, Siyuan Yu, Yang Gao, Shuai Li, Peng Yu, Aimin Hao, He Wang
chunleili.github.io/project-page-rlmuscle (IEEE Transactions on Visualization and Computer Graphics)

Motivation: in most 3D animation pipelines artists hand-tune per-muscle activations, which is time-consuming and lacks physical realism. Novelty: an RL controller trained on a 1D Hill-type surrogate automatically discovers per-muscle activations, which then drive a GPU multigrid-XPBD volumetric simulator with Hill-type fiber constraints, animating a 194-muscle full-body model at 18 ms per frame.

MGPBD: A Multigrid Accelerated Global XPBD Solver — Aug 2025
Chunlei Li, Peng Yu, Tiantian Liu, Siyuan Yu, Yuting Xiao, Shuai Li, Aimin Hao, Yang Gao, Qinping Zhao
10.1145/3721238.3730720 (SIGGRAPH)

Motivation: XPBD's Gauss-Seidel solver stalls on low-frequency errors, causing instability in high-resolution, high-stiffness simulations. Novelty: a global UA-AMG preconditioned CG solver with lazy prolongator reuse and lightweight near-kernel construction, greatly improving convergence and stability at low overhead.

Parallel Constraint Graph Partitioning and Coloring for Realtime Soft-Body Cutting — Apr 2025
Peng Yu, Ruiqi Wang, Chunlei Li, Yuxuan Li, Xiao Zhai, Yuanbo He, Hongyu Wu, Aimin Hao
10.2312/pg.20251267 (Pacific Graphics)

Motivation: cutting changes mesh topology on the fly, invalidating the precomputed constraint partitioning and coloring that parallel GPU solvers rely on. Novelty: a parallel constraint-graph re-partitioning and re-coloring scheme that restores solver parallelism instantly, enabling realtime soft-body cutting.

A Unified Particle-Based Solver for non-Newtonian Behaviors Simulation — Dec 2023
Chunlei Li, Yang Gao, Jiayi He, Tianwei Cheng, Shuai Li, Aimin Hao
10.1109/TVCG.2023.3341453 (IEEE Transactions on Visualization and Computer Graphics)

Motivation: non-Newtonian materials such as mud, dough and slime span fluid-like and solid-like behaviors that single-purpose solvers cannot capture consistently. Novelty: a unified SPH solver whose constitutive model smoothly covers viscous, elastic and plastic regimes, reproducing shear-thinning/thickening and viscoelastic effects in one framework.

Comparison between Two Eulerian-Lagrangian Methods: CFD-DEM & MPPIC on the biomass gasification in a fluidized bed Feb 2021
Chunlei Li, Qitai Eri
10.1007/s13399-021-01384-2 (Biomass Conversion and Biorefinery)

Comparative Study of Three Modified sCO2 Brayton Recompression Cycles Based on Energy and Exergy Analysis with GA Optimization Jan 2021
Chunlei Li, Qitai Eri
10.1504/IJEX.2021.115652 (International Journal of Exergy)

Multi-objective Optimization of sCO_2, sCO_2/tCO_2 Cycles Based on Energy-Exergy-Economy Balanced Analysis Apr 2022
Chunlei Li, Qitai Eri
10.1504/IJEX.2022.122308 (International Journal of Exergy)

Experience

R&D, Zeno Tech – Online June 2022 – Dec 2022
• Intern: R&D of the PBD method in the DCC software using C++.

R&D, Taichi Graphics – Beijing Feb 2023 – Sept 2023
• Intern: R&D of the PBD method. Supervised by Dr. Tiantian Liu, development of PBD solver.

R&D, Alibaba – Beijing May 2025 – Nov 2025
• R&D of the multigrid accelerated GPU-based muscle simulation in Houdini.

Awards

Top 10 Outstanding Graduate Students Candidate of Beihang University: Beihang University

Outstanding Graduate of Beihang University: Beihang University

Outstanding Graduate of NCEPU (Beijing): NCEPU (Beijing)

Skills

Languages: CET Band 6: 578, CET Band 4: 560, TOEFL: 97, GRE: 323+3.5

DCC Software: Houdini(5 years user)

Physics Engines: Newton (PR contributor), MuJoCo, PhysX